

DAX Assignment

- Create a measure named Total Customers, to calculate the number of distinct AdventureWorks customers who made a transaction.
- Create a measure named Return Rate, defined as quantity returned divided by quantity sold.

Total Customers = `DISTINCTCOUNT('AdventureWorks Sales'[CustomerKey])`

Return Rate = `DIVIDE(
 SUM('AdventureWorks Sales'[OrderQuantity]),
 SUM('AdventureWorks Returns Data'[ReturnQuantity])
)`

a) Logical Function

- Create a calculated column in the Customer Lookup table named Customer Priority:
 - If the customer is a parent and has an annual income > \$100,000, Custom Priority = Priority
 - Otherwise, Customer Priority = Standard

```
Customer Priority = IF(  
    'AdventureWorks Customer Lookup'[TotalChildren] > 0 ,  
    "Priority",  
    "Standard"  
)
```

- Create a calculated column in the Customer Lookup table named Customer income:
 - If annual income is $\geq$ \$150,000, Very High
 - If annual income is $\geq$ \$100,000, High
 - If annual income is $\geq$ \$50,000, Average
 - Otherwise, Income Level = Low

Customer Income =

```
SWITCH(  
    TRUE,  
    'AdventureWorks Customer Lookup'[AnnualIncome]  $\geq$  150000, "Very High",  
    'AdventureWorks Customer Lookup'[AnnualIncome]  $\geq$  100000, "High",  
    'AdventureWorks Customer Lookup'[AnnualIncome]  $\geq$  50000, "Average",  
    "Low"  
)
```

- Use a SWITCH function* to create another column named Education Category:
 - If EducationLevel is High School or Partial High School, Education Category = High School
 - If EducationLevel is Bachelors or Partial College, Education Category = Undergrad
 - If EducationLevel is Graduate Degree, Education Category = Graduate

```
1 Education Category = SWITCH(  
2     TRUE,  
3     OR('AdventureWorks Customer Lookup'[EducationLevel] == "High School", 'AdventureWorks Customer Lookup'[EducationLevel] == "Partial High School") , "High School",  
4     OR('AdventureWorks Customer Lookup'[EducationLevel] == "Bachelors", 'AdventureWorks Customer Lookup'[EducationLevel] == "Partial College") , "Undergrad",  
5     'AdventureWorks Customer Lookup'[EducationLevel] == "Graduate Degree", "Graduate"  
6 )
```

b) Text Function

- Update the Month Short column in the Calendar Lookup table to extract and capitalize the first 3 characters of the month name

```
Month Short = UPPER((LEFT('Calendar Lookup'[Month Name],3)))
```


- Create a new column in the Product Lookup table named SKU Category, to return any number of characters before the first hyphen in the ProductSKU column

```
= Table.AddColumn("#Replaced Value", "Custom", each Text.BeforeDelimiter([ProductSKU], "-"))
```

c) Date function

- Create a new column in the Customer Lookup table named Birth Year,to extract only the year from the BirthDate column.

```
Birth Year = YEAR('AdventureWorks Customer Lookup'[BirthDate])
```

d) Calculate Function

- Create a new measure named Bike Returns to calculate the total quantity of bikes returned

```
Bike Returns = CALCULATE(  
    Sum('AdventureWorks Returns Data'[ReturnQuantity]),  
    'AdventureWorks Product Lookup'[SKU Category] == "BK"  
)
```

- Create a matrix to show Bike Returns (values) by Start of Month (rows).
What do you notice about the volume of bike returns over time?

New Start of Month	Bike Returns
1	429
2	428
3	429
4	429
5	429
6	429
7	383
8	391
9	385
10	389
11	390
12	391
Total	429

Analysis

The returns counts stayed almost constant around the value 429 till the 6th month then we noticed a steep decline in return count to almost 383 in the 7th month then it had a positive minor bump in 8th month which led the return counts to increase to 391 but after that it got almost constant at an average of 389 count

- Create a new measure named Bike Sales to calculate the total quantity of bikes sold, and add it to the matrix. What do you notice?

```
Bike Sales = CALCULATE(  
    SUM('AdventureWorks Sales'[OrderQuantity]),  
    'AdventureWorks Product Lookup'[SKU Category] == "BK"
```

New Start of Month	Bike Returns	Bike Sales
1	429	1192
2	428	1238
3	429	1352
4	429	1450
5	429	1651
6	429	1681
7	383	753
8	391	763
9	385	771
10	389	835
11	390	879
12	391	1364
Total	429	13929

Analysis

Bike sales increased consistently from 1,192 in January to a peak of 1,681 in June, indicating strong growth in the first half of the year. Sales then dropped sharply to 753 in July, the lowest point of the year. From July onwards, sales gradually recovered, reaching 1,364 in December. Overall, the trend shows strong early growth, a significant mid-year decline, and a steady recovery toward year-end.

- Create a new measure named Bike Return Rate using either CALCULATE or DIVIDE, and add it to the matrix

Bike Return Rate = `DIVIDE([Bike Returns],[Bike Sales])`

New Start of Month	Bike Returns	Bike Sales	Bike Return Rate
1	429	1192	35.99%
2	428	1238	34.57%
3	429	1352	31.73%
4	429	1450	29.59%
5	429	1651	25.98%
6	429	1681	25.52%
7	383	753	50.86%
8	391	763	51.25%
9	385	771	49.94%
10	389	835	46.59%
11	390	879	44.37%
12	391	1364	28.67%
Total	429	13929	3.08%

e) Calculate & all function

- Create a new measure named All Returns to calculate the total number of returns, regardless of filter context

```
All Returns = CALCULATE(  
    SUM('AdventureWorks Returns Data'[ReturnQuantity]),  
    ALL()  
)
```

- Create a new measure named % of All Returns that divides Total Returns by All Returns

% of All Returns = `DIVIDE(SUM('AdventureWorks Returns Data'[ReturnQuantity]),[All Returns])`

- Create a matrix to show % of All Returns (values) by product Category Name (rows).

CategoryName	% of All Returns
Accessories	61.82%
Bikes	23.47%
Clothing	14.72%
Total	100.00%

- Which category accounts for the largest percentage of returns? The smallest?"

The category that accounts for the largest percentage of returns is Accessories and the smallest one being the Clothing.

f) Iterator Function

- Create a new measure named Total Cost that multiplies the order quantities in the Sales Data table by the product cost in the Product Lookup table, then calculates the sum

```
Total Cost = SUMX(  
    'AdventureWorks Sales',  
    'AdventureWorks Sales'[OrderQuantity] * RELATED(  
        'AdventureWorks Product Lookup'[ProductCost]  
    )  
)
```

- Create a new measure named Total Profit (revenue minus cost)

```
Total Revenue = SUMX('AdventureWorks Sales',  
    'AdventureWorks Sales'[OrderQuantity] * RELATED(  
        'AdventureWorks Product Lookup'[ProductPrice]  
    )  
)
```

```
Total Profit = [Total Revenue] - [Total Cost]
```

- Create a matrix to show Total Profit (values) by Year (rows).

Year	Total Profit
2020	\$26,01,602.33
2021	\$39,67,084.13
2022	\$38,89,028.97
Total	\$1,04,57,715.43

- How much profit has AdventureWorks earned so far in 2022

\$38,89,028.97

g) Time Intelligence Function

- Add the following measures to the model:
- Previous Month Returns
- Previous Month Orders
- Previous Month Profit
- Order Target (10% increase over previous month)
- Profit Target (10% increase over previous month)
- 90-day Rolling Profit

```
Previous Month Returns = CALCULATE(  
    SUM('AdventureWorks Returns Data'[ReturnQuantity]),  
    PREVIOUSMONTH('Calendar Lookup'[Date])  
)
```

```
Previous Month Orders = CALCULATE(  
    SUM('AdventureWorks Sales'[OrderQuantity]),  
    PREVIOUSMONTH('Calendar Lookup'[Date])  
)
```

```
Previous Month Profit = CALCULATE(  
    [Total Profit],  
    PREVIOUSMONTH('Calendar Lookup'[Date])  
)
```

Order Target = [Previous Month Orders] * 1.1

Profit Target = [Previous Month Profit] * 1.1

90-day Rolling Profit = CALCULATE(
 [Total Profit],
 DATESINPERIOD(
 'Calendar Lookup'[Date],
 MAX('Calendar Lookup'[Date]),
 -90,
 DAY
)
)